

SMART INDIA HACKATHON 2026

PlanBridge AI



Intelligent Data Capture & Schedule-Linking Layer

Problem Statement ID: 26122

Problem Statement Title: Intelligent Data Capture & Schedule-Linking Layer for Infrastructure Project Management — Real-Time Actual Progress Tracking

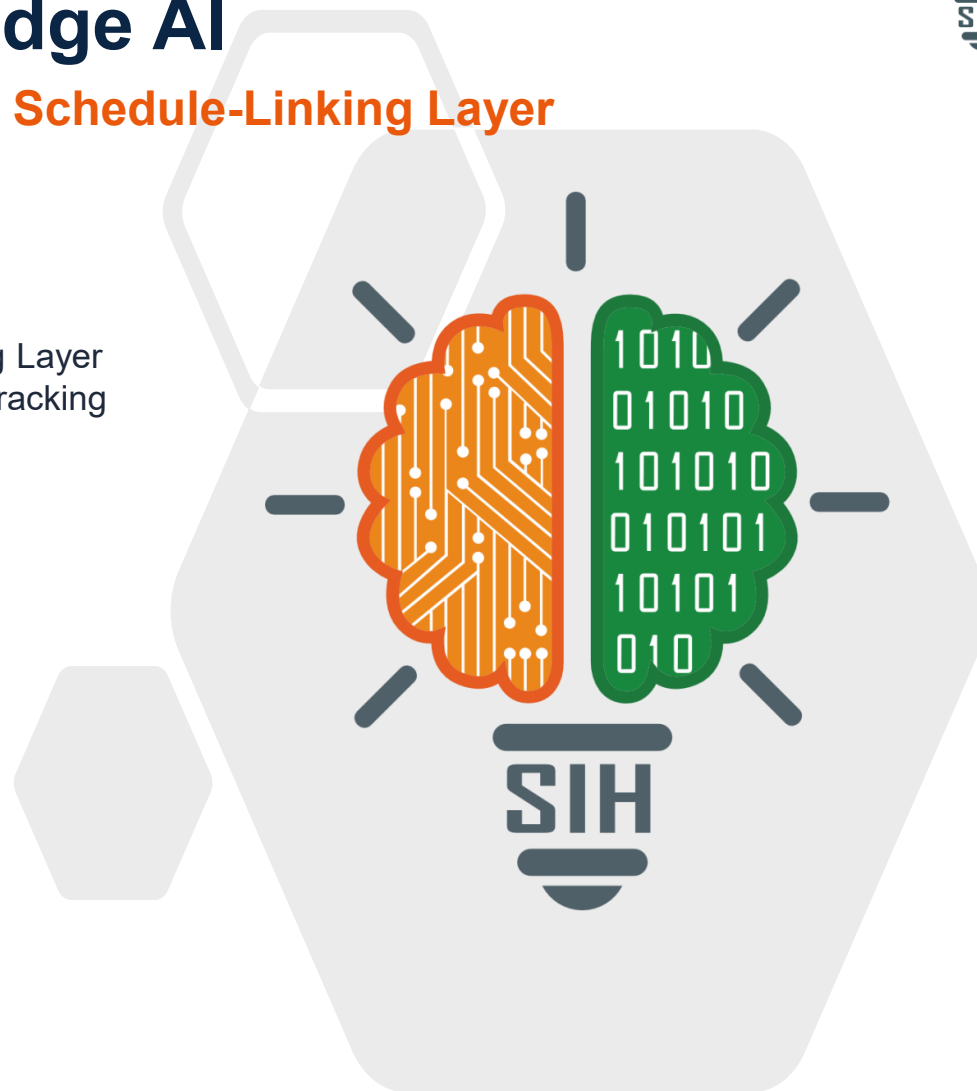
Organization: Oil India Limited

Theme: Smart Automation / Software

PS Category: Software

Team Name: [Your Team Name]

Team ID: [Your Team ID]





PROPOSED SOLUTION: PlanBridge AI



❖ 1. Proposed Solution Overview (Idea / Prototype)

- **Real-Time Planning-to-Execution Bridge:** Connects master schedules (Primavera P6 / MS Project WBS L5/L6 activities) with unstructured field reporting (daily site logs, supervisor diaries, contractor spreadsheets, conversational updates).
- **Zero-Lag Actual Progress Tracking:** Automates reconciliation guesswork, collapsing schedule lag from days/weeks to under 100 milliseconds.

2. How It Solves the Core Disconnect

- **Heterogeneous Multi-Modal Ingestion:** Parses plain-text diaries, pandas-driven tabular spreadsheets, and supervisor inputs via a conversational 'Time Agent'.
- **High-Fidelity Entity Extraction:** Extracts physical work descriptions, disciplines, dates, locations, and actions (started / in-progress / completed / blocked).
- **Semantic Vector & Candidate Matching:** Ranks schedule activities using local sentence-transformers (all-MiniLM-L6-v2), discipline compatibility, and date window proximity.

3. Innovation & Core Differentiator

- **Cross-Discipline Contradiction Detection:** Cross-checks logical precedence between concurrent disciplines (e.g. Civil reports foundation complete vs Electrical reports foundation unaligned/blocked for cable trays).
- **Human-in-the-Loop Safeguards:** High confidence (≥ 0.75) auto-updates; ambiguous matches route to a planner review queue with top-3 recommendations and an immutable audit trail.



- **1. Technology Stack & Design Decisions**

- **Backend & APIs:** Python 3.11 with FastAPI async REST endpoints and Pydantic v2 strict schemas.
- **LLM Extraction Layer:** Hosted LLM API (Gemini 1.5 Flash / OpenAI) with structured JSON output + deterministic offline structured fallback.
- **Local Vector Engine:** sentence-transformers (all-MiniLM-L6-v2) for zero-cost, offline, deterministic cosine embeddings + RapidFuzz lexical blending.
- **Database & Storage:** SQLite (Postgres-compatible schema) for resilient, zero-configuration field deployments.
- **Interactive UI/UX:** Streamlit dashboard with live Plotly analytics, conversational Time Agent, and dispute escalation center.

- **2. End-to-End Execution Pipeline & Scoring Formula**

- **Step 1 (Ingestion):** TextAdapter (free-text diaries) + SpreadsheetAdapter (pandas-driven multi-row contractor logs).
- **Step 2 (Structured Extraction):** Extracts activity description, discipline, date (YYYY-MM-DD), action type, and physical unit location.
- **Step 3 (Calibrated Scoring):** $\text{Confidence} = (0.5 \times \text{Semantic_Similarity}) + (0.3 \times \text{Discipline_Match}) + (0.2 \times \text{Date_Proximity})$.
- **Step 4 (Routing & Contradiction Scan):** $\text{Score} \geq 0.75$ auto-updates schedule; < 0.75 queues for review; ContradictionDetector scans precedence rules.
- **Step 5 (Audit Trail & Sync):** Tamper-evident SQLite audit log + CSV synchronization for Oracle Primavera P6 / SAP PMIS.

• 1. Technical Feasibility & Empirical Benchmark Results

- **Working Prototype Built & Validated:** Tested on realistic Oil India Limited Unit 3 crude pump house dataset (26 L5/L6 activities).
- **Processing Latency:** 99.04 ms average end-to-end latency (P95: 137.15 ms) from raw text ingestion to schedule update.
- **Match & Routing Accuracy:** 100.0% target activity match accuracy and 100.0% threshold routing accuracy on hand-labeled ground truth.
- **Operational Cost:** \$0.000060 USD per report (~\$0.0630 USD/month for 35 reports/day on Gemini 1.5 Flash).

2. Key Risks & Mitigation Strategies

- **Risk 1 — Non-Standard Field Slang / Typos:** Mitigation: Blended semantic embedding (all-MiniLM-L6-v2) + RapidFuzz string similarity catches colloquial site jargon.
- **Risk 2 — Contractor Reporting Disincentives:** Mitigation: Cross-discipline contradiction engine cross-verifies dependencies across contractors automatically.
- **Risk 3 — Risk of Erroneous Schedule Corruption:** Mitigation: Calibrated confidence thresholding routes ambiguous items to human engineers with top-3 candidate choices.
- **Risk 4 — Remote Oil Field Connectivity:** Mitigation: Local model execution and embedded SQLite ensure 100% offline field operational capability.

- **1. Impact on Oil India Limited & Project Controls**

- **Planning Engineers:** Saves 15–20 hours/week of manual log reading, decoding shorthand, and spreadsheet cross-referencing.
- **Site Supervisors:** Low-friction conversational logging via Time Agent removes administrative reporting burden.
- **Senior Project Leadership:** True real-time visibility into actual project health with zero status lag (from weeks to seconds).

- **2. Operational, Economic & Governance Benefits**

- **Economic Value:** Prevents contractor milestone payment disputes and arbitration claims through tamper-evident audit trails.
- **Operational Safety:** Early warning on blocked predecessor tasks prevents idle workforce standing time and site conflicts.
- **Institutional Memory Repository:** Captures actual execution durations and recurring bottlenecks to inform future capex estimates.

- **3. Commercial Viability & Scalability**

- **Broad Sector Scalability:** Extensible across all PSU infrastructure: Oil & Gas (OIL, ONGC, IOCL), Railways, Highways (NHAI), and EPC firms.
- **Enterprise Interoperability:** Standard REST APIs and CSV sync plug directly into Oracle Primavera P6 Enterprise and SAP PMIS.



- **1. Project Management Standards & Industry Best Practices**

- **PMI Practice Standard for Scheduling & WBS:** Project Management Institute (PMI) WBS Levels 1 to 6 hierarchical task decomposition and Earned Value Management (EVM).
- **Oracle Primavera P6 Enterprise Integration:** Primavera P6 REST APIs, XML/XER schedule exchange schemas, and Critical Path Method (CPM) baseline management.
- **ISO 21500 / ISO 21502:** International standards for project, programme and portfolio management — actual progress verification and quality controls.

- **2. Machine Learning & Technical Literature**

- **Sentence-BERT Architecture:** Reimers & Gurevych (2019): 'Sentence-BERT: Sentence Embeddings using Siamese BERT-Networks' (all-MiniLM-L6-v2) for dense semantic retrieval.
- **Structured LLM JSON Extraction:** Vaswani et al. (2017) 'Attention Is All You Need'; OpenAI / Google Gemini Structured JSON Schema extraction protocols.
- **Smart India Hackathon (SIH 2026):** Problem Statement ID 26122 documentation & specifications provided by Oil India Limited.